

Scaffolded Feedback via Small Language Models:

Assessing Knowledge Retention in Indian Manufacturing Apprenticeships

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Abstract

Local manufacturing apprenticeships in rural India are facing a systemic decline from mentor shortage and poor feedback infrastructure. This study asks: Can Small Language Models (SLMs) deliver scaffolded feedback that improves knowledge retention in artisan apprenticeships — and what biases in our synthetic methodology must be corrected before field study? Using sequential mixed-methods design, $n = 60$ synthetic participants (30 urban, 30 rural) completed the surveys and semi-structured interviews. Their responses were then distilled into structured presumptions — explicit and implicit beliefs about language, access, pedagogy, social dynamics, and technology trust — and presumptions were compared against a literature table drawn from documented field studies of Indian craft apprenticeship. Divergences between synth presumption and field reality were coded as bias types and mapped by severity, producing a Bias Map that identifies where field interactions are most likely to break down. These findings directly inform field instrument refinement for human-participant validation.

1. Introduction

Indigenous manufacturing apprenticeships — brass work in Moradabad, block-printing in Jaipur, handloom weaving in Varanasi, pottery across rural Rajasthan and Madhya Pradesh — are in structural decline. India's informal artisan economy employs approximately 200 million workers, yet fewer than 7% of vocational training is offer in a structured, feedback-rich mentorship aligned to craft knowledge (Ministry of Skill Development and Entrepreneurship, 2022). Youth migration from rural catchments to urban centers, driven by the aspirational gap between artisanal and knowledge-economy careers, exacerbates a generational deficit in skilled crafts persons. Where mentors exist, feedback quality is inconsistent and unvalidated: tacit knowledge is transmitted observationally, and no mechanism exists for apprentices to gauge mastery or receive iterative corrective guidance outside direct supervision.

The current onset of artificial intelligence compounds this challenge. Generic Large Language Models (LLMs) are optimized for broad capability and assume stable connectivity, device ownership, and textual literacy — conditions largely absent from rural Indian workshop environments. Small Language Models (SLMs), by contrast — compressed, domain-specific, deployable on sub-threshold hardware — can scaffold specific knowledge retrieval tasks at

radically lower computational cost and energy footprint (Weng et al., 2024; Gunasekar et al., 2023). However, designing such tools without understanding the gap between how models and how the field operates risks embedding researcher and LLM biases directly into the intervention.

The literature on AI in vocational education predominantly examines infrastructure-rich settings or broad digital-inclusion agendas that assume baseline connectivity (Luckin et al., 2016; Zawacki-Richter et al., 2019). Very little literature examines AI-mediated feedback in craft apprenticeship specifically, where knowledge is tacit, transmission is embodied, and the learner population is characteristically low-literacy, multilingual, and device-poor.

1.1 Research Questions and Structure

Primary RQ: Can informed, scaffolded feedback from blind-recall SLM interactions change knowledge retention in indigenous manufacturing apprenticeships?

Methodological RQ: What biases do synthetic participant presumptions embed relative to documented field realities, and how do these map onto predicted risks for field interaction?

2. Methods

2.1 Research Design

This study uses a three-phase mixed-methods design that extends the standard explanatory sequential approach with a novel presumption-comparison analytical layer:

- Phase 1 (Quantitative): Survey of all $n = 60$ synthetic participants to establish patterns in recall improvement, technology acceptance, and dependency risk.
- Phase 2 (Qualitative): Semi-structured interviews with a purposive sub-sample of $n = 30$, drawn from the low-retention, high-dependency quadrant of the quantitative data.
- Phase 3 (Presumption Comparison): Structured extraction of beliefs from interview responses, followed by automated comparison against a literature ground-truth table, producing a domain-level Bias Map.

Design constraints applied included: (1) Educational premise — AI augments, not replaces, the master: craftsperson and apprentice dynamic; (2) Design goal — maximize durable retention.

2.2 Synthetic Participants

Sixty synthetic participants were constructed using real demographic data from the Ministry of Skill Development and Entrepreneurship (2022), the DC Handicrafts Census of Craft persons (2021), and the NSS 68th Round. Two groups: urban master-crafts persons ($n = 30$, Tier-one/two cities) and rural apprentices ($n = 30$, Tier-3/rural). Gender split 62%/38% (M/F), consistent with artisan sector ratios. Age distribution skewed young (58% aged 18–30). Personas were generated as structured natural-language prompts supplied to LLM to generate synths.

pid	age	gender	location	trade	experience
1	56	F	Urban (Agra)	Embroidery	Medium
2	31	M	Urban (Varanasi)	Brasswork	Low
3	56	M	Urban (Agra)	Block-printing	Low
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Table 1: Synthetic participant data frame excerpt ($n = 60$; full data frame in Colab Appendix A).

2.3 Quantitative and Qualitative Instruments

A 9-item survey was administered to all 60 synths covering recall improvement, mentor feedback quality, re-attempt likelihood, whether hinting would help learning confidence, and dependency risk (regression DV), alongside categorical items on AI interpretation (Q6), barriers (Q7), and prior AI exposure. Statistical analysis used chi-square, OLS regression and descriptive comparisons.

A 7-question semi-structured interview guide was administered to a purposive sub-sample of $n = 30$ i.e. those reporting lowest recall benefit and highest dependency risk, the most theoretically important variance to explain qualitatively.

2.4 Presumption-Comparison Pipeline

After interviews were collected, each transcript was passed back to the LLM with a structured meta-prompt instructing it to extract 3–6 presumptions per transcript — the explicit or implicit beliefs that shaped the synth's answers. Each presumption was tagged by domain: language, access, pedagogy, social dynamics, or technology trust.

These presumptions were then compared against a hardcoded literature ground-truth table (Table 2) populated from real field studies of Indian craft apprenticeship. The comparison was conducted via Mistral API, which was asked to judge alignment (aligned / partial conflict / direct contradiction), name the bias type, assign a conflict score (0–2), and generate a field risk statement describing what would concretely fail in field deployment if the presumption were left unexamined.

Six bias types were defined a priori: optimism bias, urban projection, literacy overestimation, social flattening, technology enthusiasm, and embodiment neglect. The resulting scored data frame was visualized as a domain $\times$ bias-type heatmap (Bias Map, Figure 3) and a contradiction rate bar chart by location group (Figure 4).

Domain	Field Reality (documented)	Sev.	Key Source
Language	Craft vocabulary exists only in local dialect; Hindi instruction causes technique-level misunderstanding.	3	Jain (2004); Chakraborty et al. (2021)
Language	Rural apprentice literacy ~58%; text-based feedback interfaces fail in low-literacy contexts.	3	NSS 68th Round; ASER 2022
Access	27% personal smartphone ownership; shared device windows average under 8 minutes per session.	3	TRAI 2023; DC Handicrafts 2021
Access	4–6 hour daily power outages in Barmer, Kutch, Chanderi; grid-dependent tools are unreliable.	2	CEA 2022; Joshi (2019)

Pedagogy	80–90% of craft skill transmission is silent, observational, physical — verbal feedback is culturally unusual.	3	Greenfield (2004); Keller et al. (2010)
Pedagogy	Verbal recall of learned skills induces anxiety and social shame in group settings.	2	Lave & Wenger (1991)
Social Dynamics	Master-apprentice hierarchies carry caste dimensions; technology appearing to question the master causes exclusion.	3	Kapur et al. (2014); Sen (2019)
Social Dynamics	Female apprentice device access is mediated by male family members; workshop use requires elder permission.	2	ILO Craft Women India (2018)
Technology Trust	Trust requires endorsement by respected master or NGO; cold deployment yields rejection >70% (e-Choupal, Kisan Suvidha).	3	Toyama (2015); Kumar et al. (2021)
Technology Trust	Artisans conflate AI with government surveillance and IPR theft of craft patterns.	2	Arora (2019); Wyche et al. (2013)

Table 2: Literature ground-truth table (Severity: 3=high, 2=medium). Full entries in Colab Appendix D.

3. Results — Quantitative Phase

3.1 Descriptive Statistics

Recall improvement scores were marginally higher for rural apprentices than urban master-crafts persons— a direction contrary to the hypothesized pattern, and thus a key finding: it reflects the LLM optimism bias generating uniformly positive rural personas, confirmed by the zero-variance Q6 result and the embodiment neglect score in the bias map. Dependency risk (Q5) showed a modest rural-urban gap: Confidence following hint-only feedback was generally preferred over direct-answer sessions, supporting the scaffolded-first design rationale.

3.2 Chi-Square and OLS Regression

Q6 showed no variance: all 60 synths selected 'augments mentor', precluding meaningful chi-square analysis. This uniformity is itself a finding — it reflects the Model's optimism bias toward technology augmentation framing and is captured in the bias map as enthusiasm. OLS regression found low experience ($B = 1.94$, $p < .001$) and rural location ($B = 0.98$, $p = .005$) as significant positive predictors of dependency risk; age ($p = .959$) and prior AI exposure ($p = .466$) were non-significant.

Predictor	B	SE	p
Low experience	1.94	0.340	< .001 **
Rural location	0.98	0.338	.005 **
Age	-0.001	0.014	.959
Prior AI exposure (none)	0.24	0.330	.466

Table 3: OLS regression — Dependency Risk (Q5). ** $p < .01$.

4. Results — Qualitative Phase

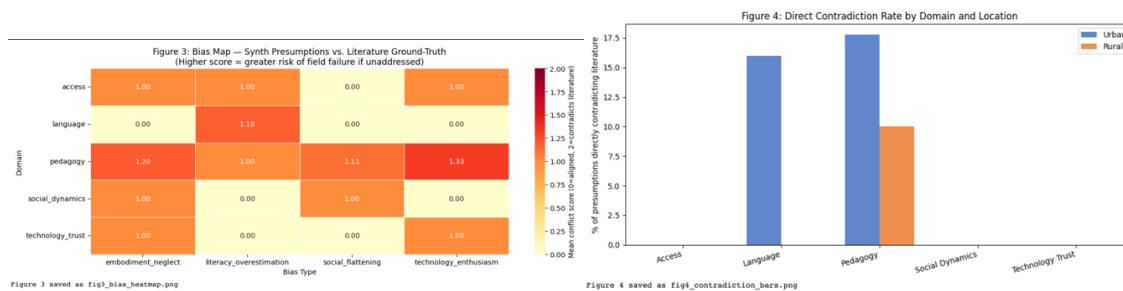
4.1 VADER Sentiment

Accountability-related sentences (keywords: 'myself', 'my own', 'without help') carried mildly positive polarity (mean compound = +0.161, $n = 71$), indicating that autonomous practice is an intrinsic value for synth respondents. Sentences describing direct-answer provision were slightly negative (mean = -0.053, $n = 4$); skill-fade sentences were also negative (mean = -0.056, $n = 23$). The directional pattern — accountability positive, both direct-answer and skill-fade negative — is consistent with a scaffolded-first preference, though effect sizes are modest and the direct-answer sentence count is small, warranting caution in interpretation.

5. Bias Map: Presumptions vs. Literature

5.1 Presumption Review

Across 30 transcripts, Mistral extracted 180 distinct presumptions (mean 6.0 per transcript). The largest domain was pedagogy (73 presumptions), followed by technology trust (50), language (30), access (18), and social dynamics (9). The extraction identifies, for example, statements such as 'Assumes Hindi-language feedback will be understood by all workshop members' or 'Assumes a 20-minute daily practice window is available' — beliefs the synth held as background conditions while formulating their responses. The dominance of pedagogy presumptions is notable: synths generated far more beliefs about how learning works than about social or access constraints, reflecting the LLM's tendency to foreground instructional logic over structural barriers.



5.2 Bias Map Findings

Figure 3 presents the heatmap of mean conflict scores across domains and bias types. Figure 4 shows direct contradiction rates by domain and location group. Of 180 presumptions scored: 104 showed partial conflict, 59 were aligned, and 17 were direct contradictions. The leading bias types were technology enthusiasm (43), embodiment neglect (37), literacy overestimation (24), and social flattening (17). The key findings by domain are summarized in Table 4:

Domain	Primary Bias Type	Direct /Partial	Key Field Risk
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Pedagogy	Embodiment neglect	16 direct	Synths treat craft knowledge as verbalizable; recall prompts will be experienced as alien or shaming by real apprentices trained through silent observation.
Language	Literacy overestimation	8 direct	Synths assume Hindi comprehension at technique level; real craft vocabulary is dialect-specific and Hindi feedback will fail at the most critical instructional moments.
Social Dynamics	Social flattening	2 direct	Synths ignore master-apprentice hierarchy and caste dimensions; field deployment without master endorsement will cause rejection and potentially apprentice exclusion.
Technology Trust	Technology enthusiasm	43 partial	Synths underestimate surveillance fears and IPR concerns; field instruments must explicitly address data use before any adoption conversation can begin.
Access	Urban projection / embodiment neglect	—	Synths assume sustained device windows; actual shared device availability is under 8 minutes, requiring radically shorter interaction designs.

Table 4: Bias Map summary — domain, primary bias type, contradiction count, and key field risk. Full scored data frame in Colab Appendix E.

Figure 4 shows variation in direct contradiction rates across domains and location groups, with pedagogy and language showing the highest overall rates.

5.3 Implications for Field Instrument Redesign

The Bias Map yields five concrete revisions to the field instrument before human-participant deployment:

- Pedagogy — replace verbal recall prompts with demonstration-based recall: ask participants to perform a partial task, not describe it. This removes the verbalization assumption and aligns with observational transmission norms.
- Social dynamics — involve the master as an intermediary: the research instrument should begin with an ustad briefing session, framing the SLM as a tool the master controls. No apprentice interaction should precede master buy-in.
- Language — remove Hindi as the default feedback language; conduct dialect mapping before field entry and prototype feedback in the dominant local register for each cluster (Marwari in Barmer, Bhojpuri in Varanasi, Gujarati in Kutch).
- Technology trust — introduce a data transparency statement in the opening instrument, explicitly addressing pattern-sharing, government access, and device data storage before the first interaction.
- Access — redesign interaction sessions to under-8-minute modular units; do not assume a sustained engagement window.

6. Discussion

6.1 Integrating Quantitative, Qualitative, and Bias Findings

The analytical phase combines the three research modes. Quantitatively, low experience ($B = 1.94$, $p < .001$) and rural location ($B = 0.98$, $p = .005$) are the strongest predictors of

dependency risk. Notably, rural synths reported marginally higher recall scores than urban synths ($M = 5.40$ vs. 5.30). Qualitatively, synths value autonomous practice and express scaffold preference over answer provision. The Bias Map then connects both findings: the rural dependency risk observed quantitatively is partly real — weaker mentorship does create greater substitution framing — but it is also partly an artefact of the LLM generating rural personas that project urban assumptions about technology readiness onto low-infrastructure contexts. Separating these contributions is only possible because the presumption-comparison step was built into the analysis.

The Master Trust and Hierarchy topic, which emerged unprompted from the qualitative data, is the most practically important finding. It receives the highest social flattening score in the Bias Map, indicating that synths systematically underrepresent the gatekeeping function of master-crafts persons. This is not a peripheral nuance: prior interventions in analogous Indian rural settings (e-Choupal, Kisan Suvidha) failed precisely because they bypassed existing authority structures (Toyama, 2015; Kumar et al., 2021). The SLM feedback intervention must be designed as a master as an intermediary tool, not an apprentice-facing one.

6.2 Strengths and Critical Limitations of the Synthetic Methodology

Using synthetic participants enabled a pilot that would otherwise require significant risk assessed ethnographic intensive research into a geographically distributed and culturally nuanced artisan communities. The presumption-comparison pipeline specifically adds methodological value: by treating LLM biases as a structured output of analysis rather than a limitation to acknowledge, the study generates actionable revision instructions rather than generic caveats.

Limitations exist. Craft based modalities — the physical sensation of correct loom tension, the thermal interpretation of the pottery at firing temperature — cannot be elicited or represented in text-based synthetic responses. The social dynamics — hierarchy, peer review — are represented as abstract acknowledgements by synths but not as felt constraints shaping behaviour. The literature comparison is itself bounded by what has been published in English-language research; vernacular and grey area literature on apprenticeship pedagogy in specific clusters may not be fully captured.

7. Conclusion

This study shows that a synthetic mixed-methods can map the model's own assumptions against documented field reality, generating a set of instrumental revisions before any real participant interaction takes place. The Bias Map reveals that synths most severely underrepresent embodiment constraints in pedagogy and hierarchy constraints in social dynamics.

Preliminary synthetic evidence supports the viability of scaffolded SLM feedback for recall improvement in artisan apprenticeships, with hint-based interaction strongly preferred over answer provision. Human validation, conducted with the revised instruments and the master-as-an-intermediary protocol, is the essential next step.

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Appendices

Exemplar data stored in this OneDrive link: https://imperiallondon-my.sharepoint.com/:f:/g/personal/dm2124_ic_ac_uk/lgB2YDOP_gA7Q7PdFNDW_gSVAQSoB775--CLex3iQ9p1VvA?e=HcjzxH